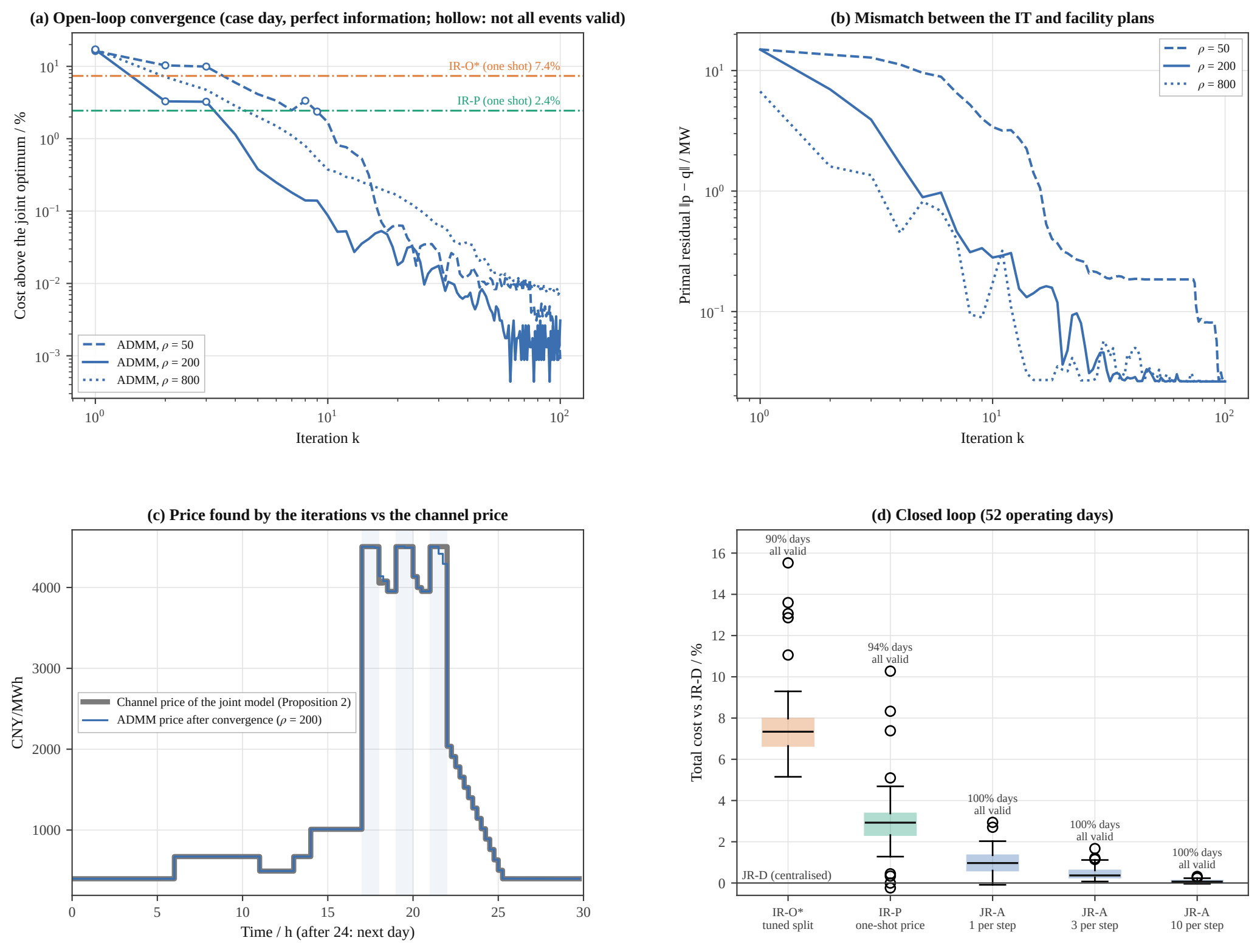
(a) Open-loop convergence (case day, perfect information; hollow: not all events valid)
Cost above the joint optimum / %
Iteration k
IR-O* (one shot) 7.4%
IR-P (one shot) 2.4%
ADMM, ρ = 50
ADMM, ρ = 200
ADMM, ρ = 800
(b) Mismatch between the IT and facility plans
Primal residual ‖p − q‖ / MW
Iteration k
ρ = 50
ρ = 200
ρ = 800
(c) Price found by the iterations vs the channel price
CNY/MWh
Time / h (after 24: next day)
Channel price of the joint model (Proposition 2)
ADMM price after convergence (ρ = 200)
(d) Closed loop (52 operating days)
Total cost vs JR-D / %
JR-D (centralised)
90% days all valid
94% days all valid
100% days all valid
100% days all valid
100% days all valid
IR-O* tuned split
IR-P one-shot price
JR-A 1 per step
JR-A 3 per step
JR-A 10 per step